

OEM Manufacturing Handbook

BioGem Lab OEM Manufacturing Handbook

Partner Operations Guide

Version: 2026-Q3

Effective Date: July 13, 2026

Classification: Confidential — Authorized Partners Only

Document Owner: BioGem Lab / Luoyang BioGem Technology Co., Ltd.

Last Updated: July 26, 2026

Section 1: Partner Onboarding

1.1 Account Setup

BioGem Lab operates a white-label, zero-inventory manufacturing model. Every partner relationship begins with the same structured onboarding sequence.

Pre-Qualification Checklist

Before first contact, verify the following about your business:

- ☐ Your brand targets the premium or luxury memorial product market
- ☐ You do not plan to resell BioGem Lab wholesale pricing to end consumers
- ☐ You have a registered business entity in your operating jurisdiction
- ☐ You understand that BioGem Lab is a B2B-only supplier and never sells to end consumers
- ☐ You can commit to a 60-day production timeline with no expedited exceptions

Initial Contact Protocol

Step	Action	Owner	Timeline
1	Send inquiry to partners@biogemlab.com or WhatsApp +1-310-437-3166	Partner	Day 0
2	Complete partner qualification questionnaire	Partner	Day 1-3
3	BioGem Lab reviews application and responds with terms	BioGem Lab	Day 3-5
4	Execute NDA and partnership agreement	Both	Day 5-10
5	Account activation and portal access	BioGem Lab	Day 10-12

Required Documentation from New Partners

- Business registration certificate
- Tax ID / VAT number
- Primary contact and authorized signatory identification
- Intended market geography (for compliance pre-check)

1.2 First Contact Standards

All BioGem Lab partner communications follow these response standards:

- Email response time: within 24 business hours
- WhatsApp response time: within 12 business hours
- Production status updates: weekly during active orders
- Invoice and shipping confirmations: same-day upon event

1.3 Sample Evaluation

New partners are eligible for one sample diamond evaluation before committing to a production order.

Sample Terms

Parameter	Specification
Sample size	0.5 ct
Carbon source	BioGem Lab internal stock (not customer material)
Deposit	USD 500, fully refundable

Parameter	Specification
Evaluation period	14 calendar days
Sample marking	Laser-inscribed “SAMPLE — NOT FOR RESALE”
Certificate	Internal QC only — not a grading certificate

Sample Evaluation Checklist

- ☐ Inspect cut quality under 10x magnification
- ☐ Verify color grade against provided specification
- ☐ Check laser inscription matches documentation
- ☐ Evaluate packaging and presentation materials
- ☐ Assess certificate layout and branding options
- ☐ Test customer-facing materials for your market

Post-Evaluation Paths

- **Satisfied:** Deposit converts to account credit or refunded within 5 business days
- **Not satisfied:** Full refund within 5 business days, sample returned to BioGem Lab at our expense
- **Proceed to order:** Deposit applied against first production order

Section 2: White-Label Service

2.1 What BioGem Lab Hides

Unless the partner explicitly requests otherwise, all customer-facing materials are delivered without BioGem Lab branding.

Elements Hidden by Default

Element	Default State	Partner Override
BioGem Lab logo	Removed	Available on request
Manufacturer name on certificate	Replaced with partner brand	N/A
Certificate template layout	Custom partner design	Standard templates available
Packaging exterior branding	Partner brand only	Co-branding on request
Origin labeling	Neutral (address only if required by law)	Partner-controlled

Element	Default State	Partner Override
Shipping documentation	Partner as shipper	BioGem Lab as shipper on request

Elements Always Present

Element	Reason
CCIC traceability code	Required for authentication
Internal serial number	Production tracking
Material source documentation	Compliance and customer trust

2.2 What the Partner Brands

The partner has full control over the following:

- Certificate issuer name and logo
- Packaging design, color, and material
- Customer-facing marketing materials
- End-consumer pricing
- Warranty and service terms
- Local customer support channels

2.3 Certificates with Partner Logo

BioGem Lab supports partner-branded certificates for the default CCIC/ILDA certificate option.

Branded Certificate Requirements

Parameter	Minimum
Order quantity for first branded certificate run	5 certificates
Artwork format	Vector (AI/EPS/PDF)
Logo size	Minimum 300 DPI for print
Turnaround for first proof	5 business days
Revision rounds included	2

Certificate Upgrade: Partner Logo on IGI/GIA

- IGI and GIA certificates can display the partner’s brand name in the “Retailer” or “Brand” field
- This is handled directly by the grading lab, not by BioGem Lab
- Partner must provide written authorization for IGI/GIA to use their brand name

Section 3: Product Specifications

3.1 Core SKU Table

BioGem Lab offers four standard SKUs. No other sizes are standard stock.

SKU	Weight	Diameter	Color	Clarity	Cut	Wholesale Price (USD)
Entry	0.5 ct	~5.0 mm	E–H	VS	Ideal	\$1,200
Core	1.0 ct	~6.5 mm	E–H	VS	Ideal	\$1,600
Premium	1.5 ct	~7.5 mm	E–H	VS	Ideal	\$2,300
Flagship	2.0 ct	~8.2 mm	E–H	VS	Ideal	\$4,999

Sales Strategy Note

The 1.0 ct SKU is the primary recommendation for partner positioning. It offers the strongest margin-to-perceived-value ratio and the fastest inventory turnover in partner channels.

3.2 Raw Material Requirements

Parameter	Specification
Minimum submission	6 g of pet hair
Recommended submission	10–20 g
Collection	Can be gathered over multiple sessions
Storage	Dry, sealed, away from direct light
Accepted sources	Pet hair (cat, dog, rabbit, horse), human hair, plant fiber
Post-purification carbon purity	99.95%+

Sample Sizing by Carbon Source

Source	Minimum	Recommended	Notes
Human / pet hair	6 g	10–20 g	1.0 ct equivalent ~6 g
Feathers	10 g	20–30 g	Lower carbon content
Plant fiber	15 g	30–50 g	High variance by species

Source	Minimum	Recommended	Notes
Solid keratin (nails, shells)	10 g	20 g	Medium extraction rate
Cremated remains	Not accepted	—	Carbon content insufficient

3.3 Non-Standard Sizes

Technically feasible but not standard:

- 0.2–0.3 ct: HPHT growth cycle is nearly identical to 0.5 ct (30–35 days vs. 35–40 days), but fixed costs (certificate, logistics) do not scale down. **Not recommended.**
- If a partner’s market positioning absolutely requires smaller sizes, contact BioGem Lab for a custom quotation.

3.4 Cut Options

Default: Round Brilliant, Ideal cut.

Available alternatives at no significant cost increase: - Cushion - Emerald - Oval - Pear - Heart

Color options: E–H only. BioGem Lab does not produce colored diamonds (blue, yellow, green, red, pink, black).

Section 4: Pricing Structure

4.1 Wholesale Price List

All prices in USD. All prices include international shipping and full insurance.

SKU	Unit Price
0.5 ct	\$1,200
1.0 ct	\$1,600
1.5 ct	\$2,300
2.0 ct	\$4,999

Pricing Discipline

- All partners receive the same base unit price.
- No published price list exists. This document is confidential.

- BioGem Lab does not provide suggested retail pricing. End-consumer pricing is entirely at the partner's discretion.

4.2 Volume Discounts

Volume incentives are delivered through the Annual Rebate Program (Section 10), not per-order discounts.

4.3 Certificate Upgrade Pricing

Certificate	Additional Cost	Added Lead Time
CCIC / ILDA (default)	Included in wholesale price	Standard
IGI	\$100–\$300 per stone	+5–7 days
GIA	\$250–\$400 per stone	+10–14 days

4.4 Custom Packaging Pricing

Item	Cost
Custom branded packaging per unit	\$12–\$18
First-order mold/setup fee	\$500 (one-time)

4.5 Multi-Stone Family Set Discount

For multiple stones from the **same carbon source** in a single order:

Stone	Price
1st	100% of unit price
2nd and onward	80% of unit price (20% discount)

Certificate upgrade fees are charged per stone and do not receive the family set discount.

Example: Three 0.5 ct stones from one pet

- Stone 1: \$1,200
- Stone 2: $\$1,200 \times 0.80 = \960
- Stone 3: $\$1,200 \times 0.80 = \960
- **Total: \$3,120** (vs. \$3,600 at list — 13.3% savings)

Section 5: Packaging Options

5.1 Option A: Branded Logistics Box

For partners selling directly to end consumers.

Contents

- ☐ Branded outer logistics box (partner logo, custom design)
- ☐ Velvet display box
- ☐ Grading certificate (CCIC default; IGI/GIA upgrade optional)
- ☐ Internal QC certificate
- ☐ Care and maintenance booklet
- ☐ Unique internal barcode

Specifications

Parameter	Detail
Minimum order for custom branding	100 units
Customization	Logo, brand color, box shape, interior material
Lead time for first branded batch	+10–14 days after design approval
Unit cost	\$12–\$18 + \$500 mold fee (first order only)

5.2 Option B: Industrial Minimal Packaging

For B2B resale, secondary processing, or white-label redistribution.

Contents

- ☐ Resealable aluminum foil pouch (moisture barrier)
- ☐ Compact diamond storage box
- ☐ Internal QC certificate
- ☐ Unique internal barcode
- ☐ CCIC traceability certificate

Specifications

Parameter	Detail
Standard unit cost	Included in wholesale price
Custom branding	Available at standard packaging rates
Shipping volume	Optimized for minimal dimensional weight
Target use case	Bulk orders, wholesale redistribution, 镶嵌前流转

5.3 Packaging Decision Matrix

Partner Type	Recommended Option	Notes
Pet aftercare / funeral home	A	End-consumer facing; presentation matters
Jewelry retailer	A or B	Depends on whether they sell loose or mounted
Brand agent / distributor	B	Cost-optimized for bulk
Custom brand (C-Type)	A	Required for brand positioning

Section 6: Order Workflow

6.1 Standard Production Timeline: 60 Days

Stage	Duration	Activity
Carbon extraction	3–5 days	Receive sample, document, begin extraction
Purification	5–7 days	Refine to 99.95%+ carbon purity
Graphitization	2–3 days	Convert to graphite for HPHT feedstock
HPHT crystal growth	18–25 days	Core production phase
Cutting & polishing	7–10 days	Transform rough to finished stone
Grading & certification	2–3 days	CCIC default; IGI/GIA if upgraded
Total	~60 days	From sample receipt to dispatch

Variance Notice

HPHT crystal growth is sensitive to temperature stability, pressure consistency, and catalyst saturation. A ± 5 day variance is normal and not grounds for compensation.

6.2 Order Workflow Checklist

Before Order Placement

- ☐ Partner has completed onboarding and account activation
- ☐ Carbon sample is collected, dried, and sealed
- ☐ Sample shipping method confirmed (FedEx/DHL/UPS recommended; avoid USPS)
- ☐ Order form completed with SKU, quantity, certificate choice, packaging choice
- ☐ Payment received (100% prepay for new partners)

During Production

- ☐ Week 1: Sample received, extraction started
- ☐ Week 2–3: Purification and graphitization
- ☐ Week 4–7: HPHT growth (no status update during this phase — normal)
- ☐ Week 8: Cutting and polishing initiated
- ☐ Week 9: Grading and certification
- ☐ Week 9+: Final QC, packaging, dispatch

After Dispatch

- ☐ Tracking number sent within 4 hours of handoff to courier
- ☐ Commercial invoice and packing list included
- ☐ Insurance certificate attached (110% of invoice value)
- ☐ Partner confirms receipt within 48 hours
- ☐ Partner inspects and accepts within 7 days

6.3 Milestone Communication Schedule

Milestone	Communication Method	Timing
Sample received	Email	Same day
Extraction complete	Email	Day 5–7
HPHT growth started	Email	Day 10–12
Cutting started	Email	Day 45–50
Grading complete	Email + certificate preview	Day 55–58
Dispatched	Email + tracking	Day 58–60

Section 7: Shipping & Logistics

7.1 Carrier and Service Level

Parameter	Specification
Primary carrier	DHL Express
Secondary carrier	FedEx International Priority
Service level	Express / Priority (not economy)
Insurance	Full value + 10%, included in wholesale price
Tracking	Door-to-door with signature required

7.2 Transit Times

Route	Typical Transit
China → United States	5–7 days
China → Japan	4–7 days
China → Europe	5–8 days
China → Australia	5–7 days

7.3 Customs Documentation

Every shipment includes:

- ☐ Commercial invoice
- ☐ Packing list
- ☐ Certificate of Origin (CO)
- ☐ Material source declaration
- ☐ HS classification statement

HS Codes

Item	HS Code	Notes
Lab-grown diamond (finished)	7104.90	China export; no Kimberley Process certificate required
Pre-purified carbon powder	3801.10.00.90	Alternative input method; no special import license needed in China
Pet hair / human hair sample	Varies by country	See Section 8 for country-specific guidance

7.4 Sample Shipping (Inbound to BioGem Lab)

Packaging Standard

- ☐ Waterproof label on outer surface
- ☐ Sealed plastic bag (primary barrier)
- ☐ Rigid mailer or padded envelope (secondary barrier)
- ☐ Declaration: “For in vitro laboratory use only”

Country-Specific Inbound Notes

Country	Status	Notes
United States	Permitted	Cat/dog hair as laboratory sample; no USDA permit required if properly declared
Japan	Permitted	Non-prohibited biological material; usually no special license
India	⚠ Complex	Consult DHL India / FedEx India and local customs broker before shipping

Section 8: Certification

8.1 Default Certification: CCIC

All standard orders include a CCIC (China Certification & Inspection Group) certificate at no additional cost.

CCIC Certificate Features

- Grading report: color, clarity, cut, carat weight
- Unique traceability QR code
- Partner brand name (if white-label activated)
- Material source documentation

8.2 Upgrade: IGI Certificate

Parameter	Detail
Cost	\$100–\$300 per stone
Added lead time	+5–7 days
Grading depth	Full 4C report with proportions diagram
Partner branding	Partner name in “Retailer” field
Recommended for	Partners whose end customers recognize IGI

8.3 Upgrade: GIA Certificate

Parameter	Detail
Cost	\$250–\$400 per stone
Added lead time	+10–14 days
Grading depth	Full GIA grading report
Partner branding	Partner name in “Report Comments” or “Retailer” field
Recommended for	Premium positioning, US market

8.4 Certificate with Partner Brand

Certificate Type	Partner Logo Possible	Process
CCIC / ILDA	Yes	BioGem Lab handles design; 5-certificate minimum
IGI	Yes	Partner submits brand authorization to IGI directly
GIA	Yes	Partner submits brand authorization to GIA directly

Section 9: Payment Terms

9.1 New Partners

Parameter	Requirement
Payment structure	100% prepayment
Payment method	International wire transfer (T/T)
Currency	USD
Order entry	Payment must clear before order enters production queue

New Partner Payment Workflow

1. Partner submits order form
2. BioGem Lab issues pro forma invoice
3. Partner initiates wire transfer
4. BioGem Lab confirms receipt (typically 2–5 business days)
5. Order enters production queue

9.2 Established Partners

Parameter	Requirement
Eligibility	12+ months of active partnership
Credit review	Required; BioGem Lab initiates
Payment structure	Net-30 (monthly settlement)
Credit limit	Determined by review

Net-30 Workflow

1. Partner submits order form (no prepayment required)
2. Order enters production queue immediately
3. Invoice issued upon dispatch
4. Payment due within 30 calendar days of invoice date
5. Late payment: production hold on new orders until balance cleared

9.3 Invoice and Record Keeping

- All invoices issued in USD
- Wire transfer fees are the sender's responsibility
- BioGem Lab provides SWIFT confirmation upon request
- Partners are responsible for their own tax compliance in their jurisdiction

Section 10: Annual Rebate Program

10.1 Rebate Tiers

Rebates are calculated on **calendar-year net sales** and issued in the first quarter of the following year.

Tier	Annual Net Sales (USD)	Rebate Rate
Tier 1	\$100,000 – \$200,000	5%
Tier 2	\$200,001 – \$400,000	10%
Tier 3	\$400,001 and above	15%

10.2 Rebate Mechanism

- Rebates are issued as **invoice credits**, not cash refunds
- Credits can be applied against any order in the rebate issuance year
- Unused credits expire 12 months after issuance
- Rebate calculation excludes shipping, insurance, taxes, and certificate upgrade fees

10.3 Rebate Example

Partner achieves \$250,000 in net annual sales:

- Tier 2 applies: 10% rebate
- Rebate credit: $\$250,000 \times 10\% = \textbf{\$25,000}$
- Issued as credit note in Q1 of following year
- Partner applies credit against new orders over the next 12 months

Section 11: Sample Policy

11.1 Eligibility

Sample evaluation is available to:

- B-Class partners (jewelry/gift retailers)
- C-Class partners (brand agents)

A-Class partners (pet aftercare) may request samples on a case-by-case basis.

11.2 Sample Terms

Parameter	Specification
Sample quantity	1 stone
Sample size	0.5 ct

Parameter	Specification
Carbon source	BioGem Lab internal stock
Deposit	USD 500
Evaluation period	14 days
Refund policy	Full refund if not satisfied; sample returned
Laser marking	“SAMPLE — NOT FOR RESALE”

11.3 Sample to Order Conversion

- Satisfied partners may retain the sample as a display piece
- The \$500 deposit is applied against the first production order
- If no order is placed within 90 days, the deposit is refunded and the sample is recalled

Section 12: Support Contacts

12.1 Primary Contact Channels

Channel	Address / Number	Purpose	Response Time
Email	partners@biogemlab.com	General inquiries, orders, documentation	24 business hours
WhatsApp	+1-310-437-3166	Urgent issues, quick questions	12 business hours
WeChat	Available upon request	Partners in Greater China region	24 business hours

12.2 Escalation Path

Level	Trigger	Contact
Standard	All routine matters	partners@biogemlab.com
Urgent	Production delay >5 days past variance	Same + WhatsApp
Critical	Quality dispute, customs hold, lost shipment	Direct to operations manager

12.3 Operations Entity

Luoyang BioGem Technology Co., Ltd.

11F, R&D Building, Zhongke Information Industry Park
Yibin District, Luoyang, Henan, China

Business Registration: 91410300MA9KJT0M47

12.4 Hours of Operation

BioGem Lab operates on China Standard Time (UTC+8).

Day	Hours (CST)	Corresponding EST	Corresponding PST
Monday– Friday	09:00 – 18:00	20:00 – 05:00 (prev day)	17:00 – 02:00 (prev day)
Saturday	09:00 – 12:00	20:00 – 23:00 (Fri)	17:00 – 20:00 (Fri)
Sunday	Closed	—	—

Appendix A: Compliance and Legal

A.1 Confidentiality

This handbook and all associated pricing are proprietary and confidential. Unauthorized disclosure, distribution, or reproduction is prohibited. Partners may not share BioGem Lab wholesale pricing or cost structures with end consumers or competitors.

A.2 Quote Validity

All pricing in this handbook is valid for 90 days from the date of issue. BioGem Lab reserves the right to adjust pricing with 30 days' written notice.

A.3 Return Policy

- Returns accepted only for manufacturing defects
- Transport damage covered by shipment insurance
- Color deviation: if final color grade deviates by more than one full grade from specification, BioGem Lab will remake at no charge (same 60-day timeline)

A.4 No Expedited Service

As of July 1, 2026, BioGem Lab does not offer expedited production. All orders follow the standard 60-day timeline. Partners should communicate this clearly to their end customers.

A.5 Geographical Scope

BioGem Lab currently serves partners in 14+ countries. New market entries are evaluated on a case-by-case basis.

Appendix B: Partner Classification

Class	Description	Recommended First Order	Notes
A — Pet Aftercare	Pet cremation, funeral homes, memorial services	10 stones (0.5 ct and 1.0 ct split)	End-consumer facing; highest service expectations
B — Jewelry / Gift	Independent jewelers, premium pet retailers	10 stones, full prepayment	Mix of loose and mounted sales
C — Brand Agent	Independent brand seeking white-label OEM	50 stones with custom packaging	Full white-label; highest customization

Appendix C: Document Control

Version	Date	Changes
2026-Q3	Jul 13, 2026	Initial release based on B2B wholesale quote PDF
2026-Q3.1	Jul 26, 2026	Added OEM Manufacturing Handbook format

End of Document